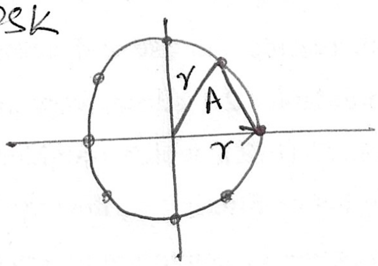
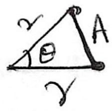
problem 1

8 PSK



'r' in terms of A.



$\theta = \frac{\pi}{4}$

use cosine rule

$$r^2 + r^2 - 2r^2\cos\theta = A^2$$

$$2r^2 - 2r^2\frac{1}{\sqrt{2}} = A^2$$

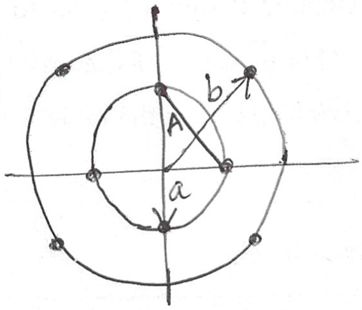
$$r = \frac{A}{\sqrt{2-\sqrt{2}}} = 1.306A$$

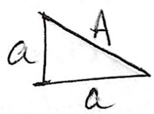
$r = 1.306A$

$E_{avg} = p(s_i)\sum_{i=1}^{K} d_i^2$

$E_{avg} = \frac{1}{8} 8r^2 = 1.707A^2 \rightarrow ①$

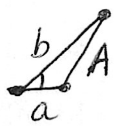
8 QAM





$A = \sqrt{2}a$

$a = \frac{A}{\sqrt{2}} = 0.707A$



use cosine rule

$$A^2 = a^2 + b^2 - 2ab\cos 45°$$

$$A^2 = a^2 + b^2 - 2ab\frac{1}{\sqrt{2}}$$

$$A^2 = \left(\frac{A}{\sqrt{2}}\right)^2 + b^2 \, 2\frac{A}{\sqrt{2}} b \frac{1}{\sqrt{2}}$$

$$b = \frac{A}{2}(1+\sqrt{3})$$

$$b = 1.366A$$

$E_{avg} = \frac{1}{8}[4a^2 + 4b^2]$

$E_{avg} = 1.183A^2 \rightarrow ②$

From Eq^n ① & ②, 8-QAM is more Energy Efficient.

$$\text{Power Advantage} = 10\log_{10}\left(\frac{1.707A^2}{1.183A^2}\right)$$

$$P_{8PSK} = (1.592\,dB)\, P_{8-QAM}$$

8-PSK

$$P_{e\ \text{optimistic}} = p(s_i)\ \text{Number of symbols} \times \text{Immediate neighbouring symbols}\ \frac{1}{2}\,\text{Erfc}\left(\frac{d_{min}}{2\sqrt{N_0}}\right)$$

$$= \frac{1}{8}\ 8 \times 2 \times \frac{1}{2}\,\text{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

$$= \text{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

$$P_{e\ \text{pessimistic}} = p(s_i)\ \text{Number of symbols} \times (M-1) \times \frac{1}{2}\,\text{Erfc}\left(\frac{d_{min}}{2\sqrt{N_0}}\right)$$

$$= \frac{8}{8} \times 7 \times \frac{1}{2}\,\text{Erfc}\left(A/2\sqrt{N_0}\right)$$

$$= 3.5\ \text{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

8-QAM

$$P_{e\ \text{optimistic}} = \frac{1}{8}\left[(4 \times 2) + (4 \times 4)\right]\frac{1}{2}\,\text{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

$$= 1.5\ \text{Erfc}\left(A/2\sqrt{N_0}\right)$$

$$P_{e\ \text{pessimistic}} = 3.5\ \text{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

2 Problem 2

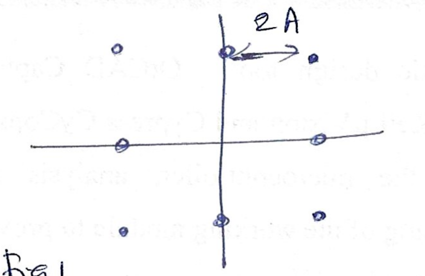


Fig 1

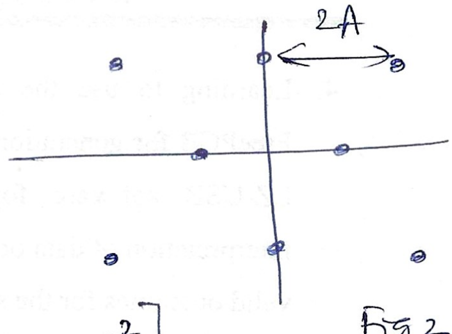


Fig 2

Fig 1: $E_{bavg} = \frac{1}{8}\left[4 \times (2A)^2 + 4 \times (2\sqrt{2}A)^2\right]$

$= 6A^2$

Fig 2: $E_{bavg} = \frac{1}{8}\left[2A^2 + 2(A\sqrt{3})^2 + 4(A\sqrt{7})^2\right]$

$= 4.5A^2$

Fig 2, 8-QAM has lower Avg energy compared to Fig 1, 8-QAM.

Fig 1

P_e optimistic $= \frac{1}{8}\left[4 \times 2 + 2 \times 2 + 2 \times 2\right] \frac{1}{2} \operatorname{Erfc}\left(\frac{2A}{2\sqrt{N_0}}\right)$

$= \operatorname{Erfc}\left(A/2\sqrt{N_0}\right)$

P_e pessimistic $= 3.5 \operatorname{Erfc}\left(A/\sqrt{N_0}\right)$

Fig 2

P_e optimistic $= \frac{1}{8}\left[4 \times 2 + 2 \times 4 + 2 \times 5\right] \frac{1}{2} \operatorname{Erfc}\left(2A/2\sqrt{N_0}\right)$

$= 1.625 \operatorname{Erfc}\left(A/\sqrt{N_0}\right)$

P_e pessimistic $= 3.5 \operatorname{Erfc}\left(A/\sqrt{N_0}\right)$

Problem 3

$$E_{avg} = E_{savg} = \frac{1}{16}\left[4 \times (\sqrt{2}A)^2 + 8 \times (\sqrt{10}A)^2 + 4 (\sqrt{18}A)^2\right]$$

$$E_{savg} = 10A^2$$

$$E_{bavg} = \frac{E_{savg}}{\log_2 M} = 2.5A^2$$

$$A = \sqrt{\frac{E_{bavg}}{2.5}}$$

$$\text{QAM } P_{e\,optimestic} = \frac{1}{16}\left[4\times4 + 4\times2 + 8\times3\right]\frac{1}{2}\,\text{Erfc}\left(\frac{2A}{2\sqrt{N_0}}\right)$$

$$= 3\,\text{Erfc}\left(A/\sqrt{N_0}\right)$$

$$= 3\,\text{Erfc}\left(\sqrt{\frac{E_{bavg}}{2.5N_0}}\right)$$

$$= 3\,\text{Erfc}\left(\sqrt{\frac{0.4\,E_{bavg}}{N_0}}\right)$$

$$\text{PAM } P_{e\,optimistic} = \text{Erfc}\left[\sqrt{\frac{3\log_2 M \;E_{bavg}}{(M^2-1)N_0}}\right]$$

$$= \text{Erfc}\left(\sqrt{0.047\frac{E_{bavg}}{N_0}}\right)$$

Efficiency $= \eta$

$$\frac{\eta_{QAM}}{\eta_{PAM}} = \frac{0.4}{0.047} = 8.5 = 9.3\text{dB}$$

QAM is more power Efficient.

$$\text{QAM } P_{e\,pessimistic} = 7.5\,\text{Erfc}\left(\frac{A}{\sqrt{N_0}}\right)$$

Problem 4

Fig 1: 8-QAM

$$E_{savg} = \frac{1}{8}\left[2(A/2)^2 + 2\left(\frac{\sqrt{3}}{2}A\right)^2 + 4\left(\frac{\sqrt{7}}{2}A\right)^2\right]$$

$$= 1.125\,A^2$$

$$P_{e\ \text{optimistic}} = \frac{1}{8}\left[4\times 2 + 2\times 5 * 2\times 4\right]\frac{1}{2}\,\text{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

$$= 1.625\,\text{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

$$P_{e\ \text{pessimistic}} = \frac{1}{8}\times 8\times 7\times\frac{1}{2}\,\text{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

$$= 3.5\,\text{Erfc}\left(A/2\sqrt{N_0}\right)$$

Fig 2: 8-QAM

$$E_{savg} = \frac{1}{8}\left[4\,(A/\sqrt{2})^2 + 4\left(\frac{1+\sqrt{3}}{2}A\right)^2\right]$$

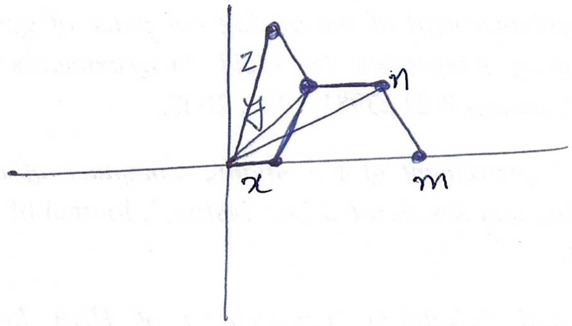
$$= 1.183\,A^2$$

$$P_{e\ \text{optimistic}} = \frac{1}{8}\left[4\times 2 + 4\times 4\right]\frac{1}{2}\,\text{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

$$= 1.5\,\text{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

$$P_{e\ \text{pessimistic}} = 3.5\,\text{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

Problem 5



$$x = A/2 \ , \quad y^2 = x^2 + A^2 - 2Ax \cos(120^\circ)$$

$$y = \frac{\sqrt{7}}{2} A$$

$$z^2 = (\sqrt{3}A) + (A/2)^2 = \frac{\sqrt{13}}{2} A$$

$$m = 2A + A/2 = \frac{5}{2} A$$

$$n = \left(\frac{\sqrt{3}}{2} A\right)^2 + (2A)^2 = \frac{\sqrt{19}}{2} A$$

$$E_{s\,avg} = \frac{1}{16}\left[2x^2 + 4y^2 + 4z^2 + 2m^2 + 4n^2\right]$$

$$= 3.25 A^2$$

$$P_{e\ \text{optimistic}} = \frac{1}{16}\left[2 \times 2 + 8 \times 2 + 6 \times 3\right] \frac{1}{2} \operatorname{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

$$= 1.1875 \operatorname{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

$$P_{e\ \text{pessimistic}} = 7.5 \operatorname{Erfc}\left(\frac{A}{2\sqrt{N_0}}\right)$$

Example 6

A binary PSK is used for a transmission over AWGN with a PSD of $\frac{1}{2} N_0 = 10^{-10}$ W/Hz. The transmitted signal Energy $E_b = \frac{1}{2} A^2 T$. where T is the bit interval and A is the signal amplitude. Determine the signal amplitude required to achieve P_e of 10^{-6} when the data rate is 10kbps, 100kbps and 1Mbps.

Soln :-

$$P_e = \frac{1}{2} \operatorname{Erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

$$10^{-6} = \frac{1}{2} \operatorname{Erfc}\left(\sqrt{\frac{A^2 T}{2N_0}}\right)$$

use $\operatorname{Erfc}(x) \approx \frac{\exp(-x^2)}{\sqrt{\pi}}$

$$\ln\left(2 \times \sqrt{\pi} \times 10^{-6}\right) = \frac{A^2 T}{2N_0}$$

$$12.54 = \frac{A^2 T}{2N_0}$$

$$A = \sqrt{12.54 \times 2N_0 \times R}$$

at $R = 10$ Kbps , $A = 7.08$ mV

$R = 100$ kbps , $A = 22.39$ mV

$R = 1$ Mbps , $A = 70.82$ mV

Example 7

Consider a digital communication system that transmits information via QAM over a voice-band telephone channel at a rate of 2400 symbols/sec. The additive noise is assumed to be Gaussian.

Determine
1) $\frac{E_b}{N_o}$ required to achieve $P_e = 10^{-5}$ at 4800 bits/sec
2) Repeat the analysis for 9600 bits/sec and 19200 bits/sec
3) What is the conclusion from the result ?

Soln :- (1) $K = \frac{R_b}{R_s} = \frac{4800}{2400} = 2$

$M = 2^k$, 4-QAM is used.

$$P_{e\,M\text{-}QAM} = 2\left(1 - \frac{1}{\sqrt{M}}\right) \text{Erfc}\left(\sqrt{\frac{3\log_2\sqrt{M}}{(M-1)} \frac{E_{bavg}}{N_0}}\right)$$

$$10^{-5} = 2 \times 0.5 \text{ Erfc}\left(\sqrt{\frac{3}{3}\frac{E_b}{N_o}}\right)$$

$$10^{-5} = \text{Erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

$$\frac{e^{-E_b/N_0}}{\sqrt{\pi}} = 10^{-5}$$

$$\frac{E_b}{N_0} = 10.94 = 10.39 \text{ dB}$$

(2) $R_b = 9600$, $k = \frac{9600}{2400} = 4$

$M = 2^k = 16$-QAM

$$P_e = \frac{3}{2} \text{Erfc}\left(\sqrt{\frac{6}{15}\frac{E_b}{N_0}}\right)$$

$$\frac{E_b}{N_0} = 27.35 = 14.36 \text{ dB}$$

When $R_b = 19{,}200$, $k = 8$

$M = 256$ QAM

$$P_e = \frac{15}{8} \, \text{Erfc}\left(\sqrt{\frac{12}{255} \frac{E_b}{N_0}}\right)$$

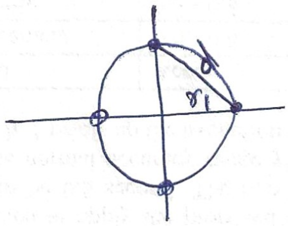
$$\frac{E_b}{N_0} = 232.47 = 23.66 \text{ dB}$$

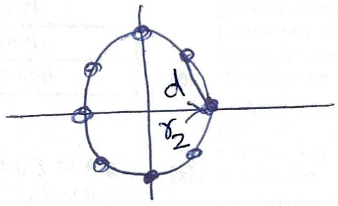
3 As $M \uparrow$ $\left(\frac{E_b}{N_0}\right)$ also increases.

Example-8

Consider a 4-PSK and 8-PSK as shown in the fig below. The minimum distance b/w two points is d. Determine the additional transmitted energy required in the 8-PSK signal to achieve the same Pe as the QPSK signal at high SNR, when Pe is determined by optimistic method.

Soln :-





$$r_1^2 + r_1^2 = d^2$$

$$r_1 = \frac{d}{\sqrt{2}}$$

$$P_{avg} = 4 \cdot \frac{1}{4} r_1^2 = \frac{d^2}{2}$$

$$2r_2^2 - 2r_2^2 \cos 45^\circ = d^2$$

$$r_2 = \frac{d}{\sqrt{2-\sqrt{2}}}$$

$$P_{avg} = 8 \frac{1}{8} r_2^2$$

$$= \frac{d^2}{2-\sqrt{2}}$$

$$10\log_{10}\left[\frac{P_{avg}\ (8\,PSK)}{P_{avg}\ QPSK}\right] \Rightarrow \underline{P_{avg\ 8\text{-}PSK} = 3.44\ P_{avg\ QPSK}}$$

$$P_{avg\ 8PSK}\big|_{dB} = 5.33\big|_{dB} + P_{avg\ QPSK}\big|_{dB}$$

Example - 9

Digital Information is to be transmitted by carrier modulation through AWGN channel with $B = 100 KHz$ and $N_0 = 10^{-10}$ W/Hz. Determine the maximum rate that can be transmitted through the channel for QPSK, BFSK and 4-FSK which is detected non-coherently.

Sol^n :-

QPSK

$$W = \frac{R_b}{\log_2 M}$$

$$R_b = (100K)(2)$$

$$R_b = 200 \text{ kbps}$$

BFSK

$$W = \frac{M R_b}{2 \log_2 M}$$

$$R_b = \frac{2 \log_2 M \cdot W}{M}$$

$$= \frac{2W}{2} = W$$

$$R_b = 100 \text{kbps}$$

4-FSK

$$W = \frac{4 R_b}{2 \log_2 4}$$

$$W = R_b$$

$$R_b = 100 \text{Kbps}$$